

An Image is Worth 16×16 Words

Transformers for Image Recognition at Scale

Dosovitskiy et al., ICLR 2021

Team ViTs

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Motivation: Why Transformers for Vision?

Transformers in NLP:

- Transformers revolutionized NLP
- Excellent scalability
- Self Attention mechanism

Computer Vision:

- CNNs dominate image recognition
- Limited use of pure attention
- Strong inductive biases

Key Question

Can we apply pure Transformers directly to images?

Challenge

Images are 2D, Transformers expect 1D sequences

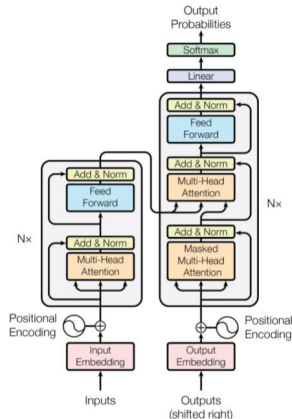


Figure: Attention Is All You Need (Vaswani et al. 2017)

Core Idea: Treating Images as Sequences

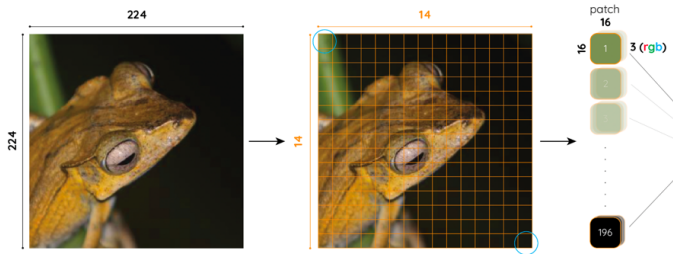


Figure: Divided into 196(14×14) patches of size $3 \times 16 \times 16$

Core Idea:

- Split an image into fixed size patches
- Treat it as embedding in NLP
- Use it as an input for traditional transformer encoder

Vision Transformer Architecture

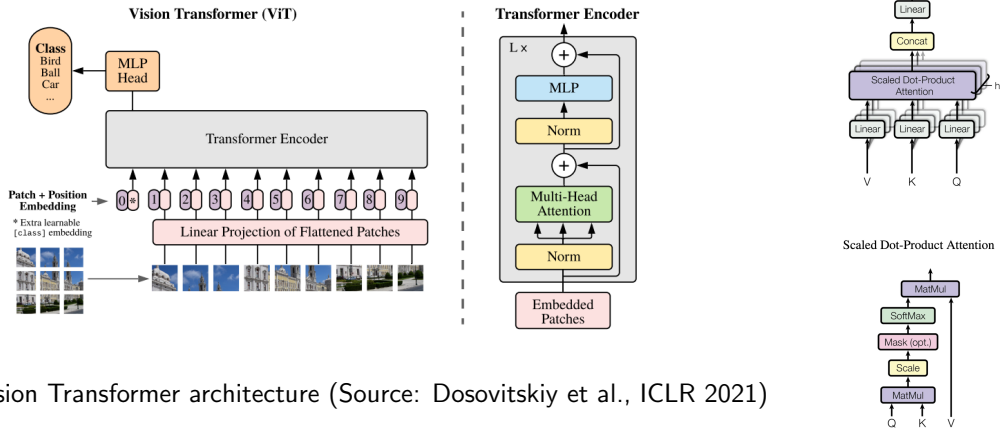
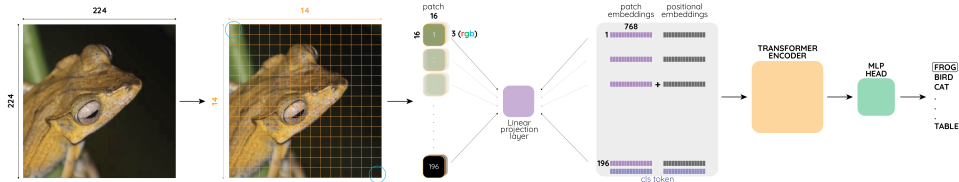


Figure: Vision Transformer architecture (Source: Dosovitskiy et al., ICLR 2021)

Vision Transformer Architecture cont.



Mathematics Behind ViT

$$\begin{aligned} \mathbf{z}_0 &= [\mathbf{x}_{\text{class}}; \mathbf{x}_p^1 \mathbf{E}; \mathbf{x}_p^2 \mathbf{E}; \cdots; \mathbf{x}_p^N \mathbf{E}] + \mathbf{E}_{pos}, & \mathbf{E} &\in \mathbb{R}^{(P^2 \cdot C) \times D}, \mathbf{E}_{pos} \in \mathbb{R}^{(N+1) \times D} \\ \mathbf{z}'_\ell &= \text{MSA}(\text{LN}(\mathbf{z}_{\ell-1})) + \mathbf{z}_{\ell-1}, & \ell &= 1 \dots L \\ \mathbf{z}_\ell &= \text{MLP}(\text{LN}(\mathbf{z}'_\ell)) + \mathbf{z}'_\ell, & \ell &= 1 \dots L \\ \mathbf{y} &= \text{LN}(\mathbf{z}_L^0) \end{aligned}$$

Vision Transformer Variants

Model	Layers	Hidden size D	MLP size	Heads	Params
ViT-Base	12	768	3072	12	86M
ViT-Large	24	1024	4096	16	307M
ViT-Huge	32	1280	5120	16	632M

Pre training : JFT300M (224 × 224)

Vision Transformer: Components and Insights

Main Components

- **Image Patching** – Split image into fixed size patches
- **Linear Projection** – Flatten patches into vectors (embeddings).
- **Positional Encoding** – Add patch location info.
- **Class Token (CLS)** – Special learnable token for classification.
- **Multi Head Self Attention** – Each patch attends to all others (global context).
- **MLP** – Nonlinear transformation of patch features.
- **Residual + Norm** – Stabilize and speed up training.
- **Classification Head** – Uses [CLS] token to predict final class label.

What ViT Learns

- **Patch Embeddings** – Learns edges, textures, and simple shapes inside patches (like CNN filters).
- **Positional Embeddings** – Learns patch locations in the image.
- **Attention** – Learns how each image patch relates to every other patch
- **MLP** – Early layers focus on low level details, deeper layers capture high level semantics.
- **Class Token Representation** – The [CLS] token learns to summarize the whole image for classification.

Vision Transformer: Key Takeaways

- ① **Simplicity:** Pure Transformer applied directly to image patches
- ② **Minimal inductive bias:** Relies on data
- ③ **Efficiency:** Less computational resources for pre training

Impact

ViT demonstrates that *reliance on CNNs is not necessary* and can apply Attention and Transformer for vision tasks

Thank you for your attention!